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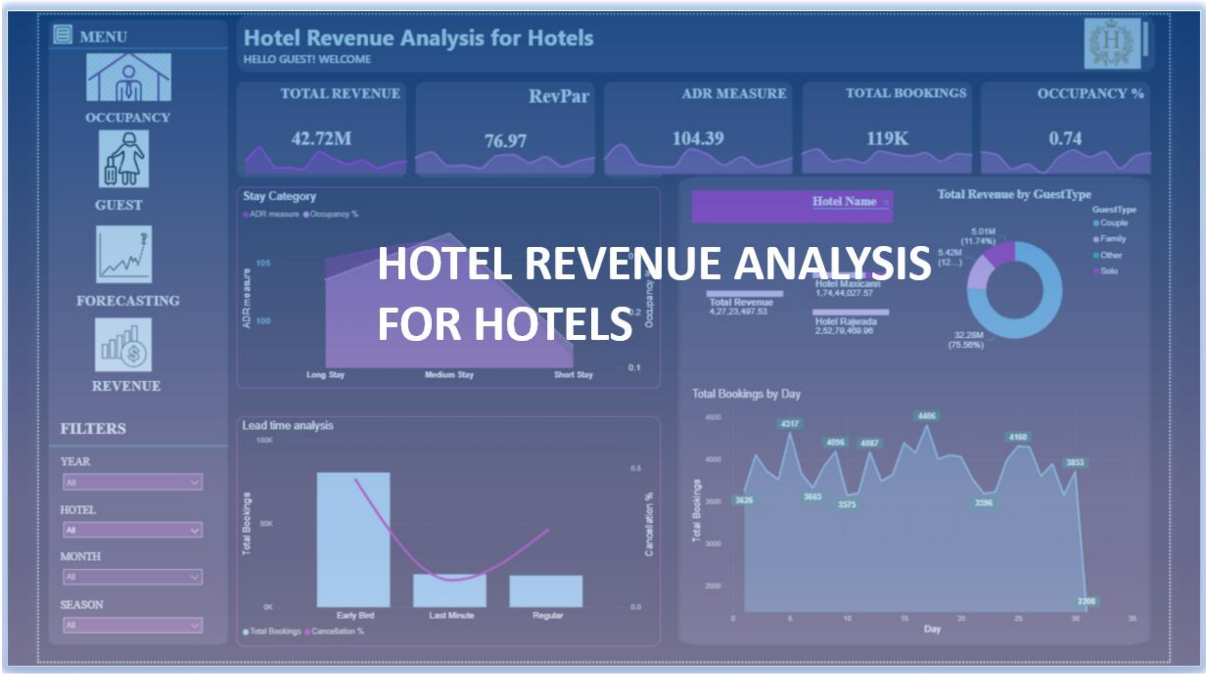
Hotel Revenue Analysis Dashboard using Power BI



Project Statement

The hospitality industry faces challenges in maximizing revenue due to fluctuating demand, seasonal variations, booking cancellations, and varying guest behavior. Simply tracking bookings is not enough; hotels must analyze revenue drivers, pricing effectiveness, upselling potential, and cancellation risks to improve profitability. This project develops an advanced Hotel Revenue Strategy Dashboard using Power BI to analyze occupancy trends, Average Daily Rate (ADR), revenue distribution, deposit policies, cancellation impact, and upsell potential.

Outcomes

• Interactive occupancy rate and booking trend dashboards
• ADR (Average Daily Rate) and RevPAR (Revenue Per Available Room) performance metrics
• Customer segmentation based on stay purpose and guest demographics
• Forecasting of occupancy and revenue using historical data trends
• Branch-level and hotel-type performance comparisons
• Strategic dashboard to support pricing and marketing decisions
• Identification of upselling opportunities to increase revenue per guest
• Scalable data model applicable across multiple hotel chains and geographical regions

Software Requirements

- Microsoft Power BI Desktop
- Microsoft Excel (for data cleaning and preparation)
- Windows Operating System
- Power BI DAX for calculated measures
- Power Query for data transformation

Hardware Requirements

Minimum:

- Processor: Intel i3 or above
- RAM: 4 GB minimum (8 GB recommended)
- Hard Disk: 5 GB free space
- Display: 1366 × 768 resolution or higher

Recommended:

- Intel i5/i7 processor
- 8–16 GB RAM for better performance

Modules to be Implemented:

Module 1: Occupancy & Revenue Analysis

This module focuses on measuring overall hotel performance in terms of revenue, bookings, pricing efficiency, and occupancy trends.

• Display Total Revenue, RevPAR, ADR, Total Bookings, and Occupancy % using KPI cards
• Analyze stay categories (Long, Medium, Short stay) against ADR and Occupancy %
• Evaluate lead time categories (Early Bird, Regular, Last Minute) with cancellation impact
• Visualize total bookings trend by day
• Compare revenue distribution by guest type (Couple, Family, Solo, Other)
• Analyze performance by hotel branch (Hotel Maxicann & Hotel Rajwada)

Module 2: Guest Analysis Module

This module focuses on understanding guest behavior, stay patterns, and booking sources to identify upselling and revenue opportunities.

• Analyze total bookings by guest type using tree map
• Compare average stay duration by guest segment
• Evaluate revenue contribution by guest type and booking source (Direct vs OTA)
• Analyze ADR performance across guest categories
• Visualize total bookings by country using geographic map

Module 3: Forecasting & Cancellation Analysis

This module focuses on analyzing booking cancellations, revenue loss, and demand forecasting using historical trends. It helps management understand revenue risk and improve booking policies.

• Analyze cancellation patterns across lead time categories
• Measure revenue lost due to booking cancellations
• Identify high-risk booking segments
• Study seasonal variation in cancellation percentage
• Forecast occupancy trends using rolling average method
• Support strategic decisions related to deposit and refund policies

Module 4: Revenue Strategy & Upselling Optimization

This module focuses on identifying revenue drivers, seasonal pricing impact, room performance, meal package contribution, and upselling opportunities. It supports strategic decision-making to maximize revenue per guest.

• Analyze seasonal pricing impact using ADR
• Identify high-performing room categories
• Evaluate revenue contribution by meal type
• Measure revenue distribution by upsell potential
• Recommend strategies to improve revenue per booking

Methodology

Data Collection

The dataset includes:
• Booking information
• Guest details
• Room category data
• Revenue and pricing details
• Deposit and cancellation information
• Seasonal and date-based records

Data Cleaning & Preparation

• Removed duplicates and null values
• Standardized date formats
• Categorized lead time into Early Bird, Regular, and Last Minute
• Created stay categories (Short, Medium, Long Stay)
• Segmented guests by upsell potential
• Classified deposit types

Data Modeling

• Fact Table: Booking & Revenue Data
• Dimension Tables: Date, Hotel, Guest Type, Room Type, Meal Type

KPI Development (Using DAX)

Occupancy %, Average Daily Rate (ADR), RevPAR (Revenue Per Available Room), Total Revenue, Lost Revenue

Dashboard Design & Visualization

• Occupancy & Revenue Analysis

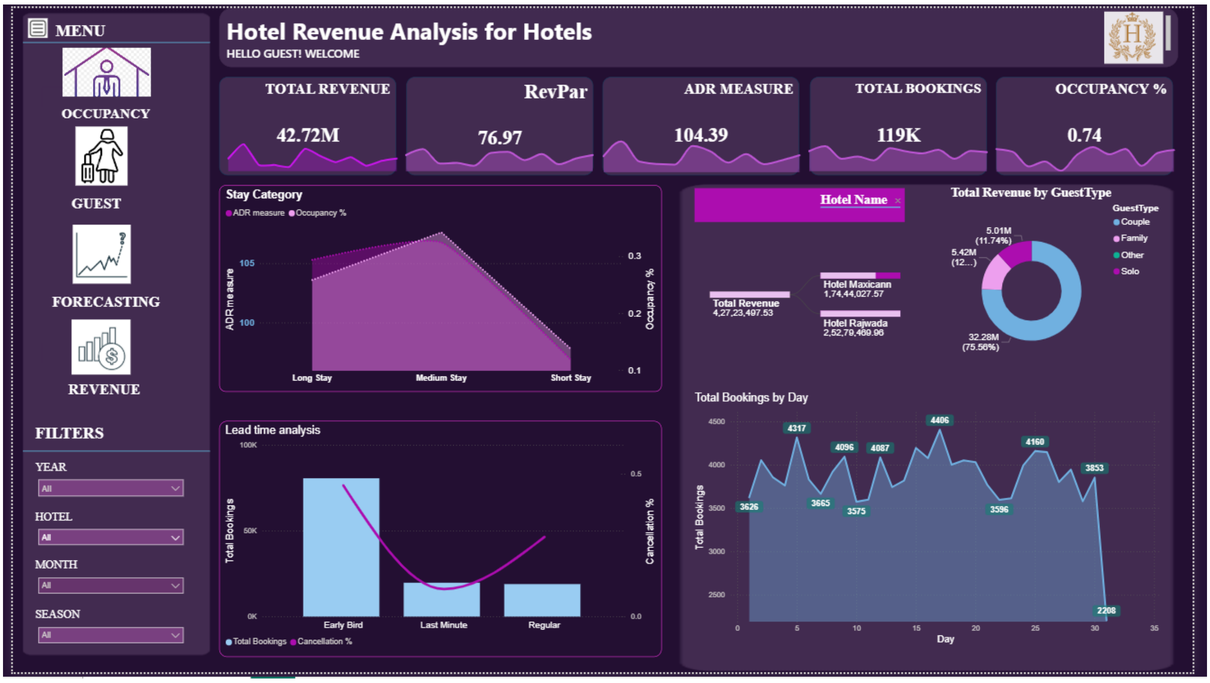


Fig 1 : Occupancy and Revenue Analysis Dashboard

• Guest Analysis

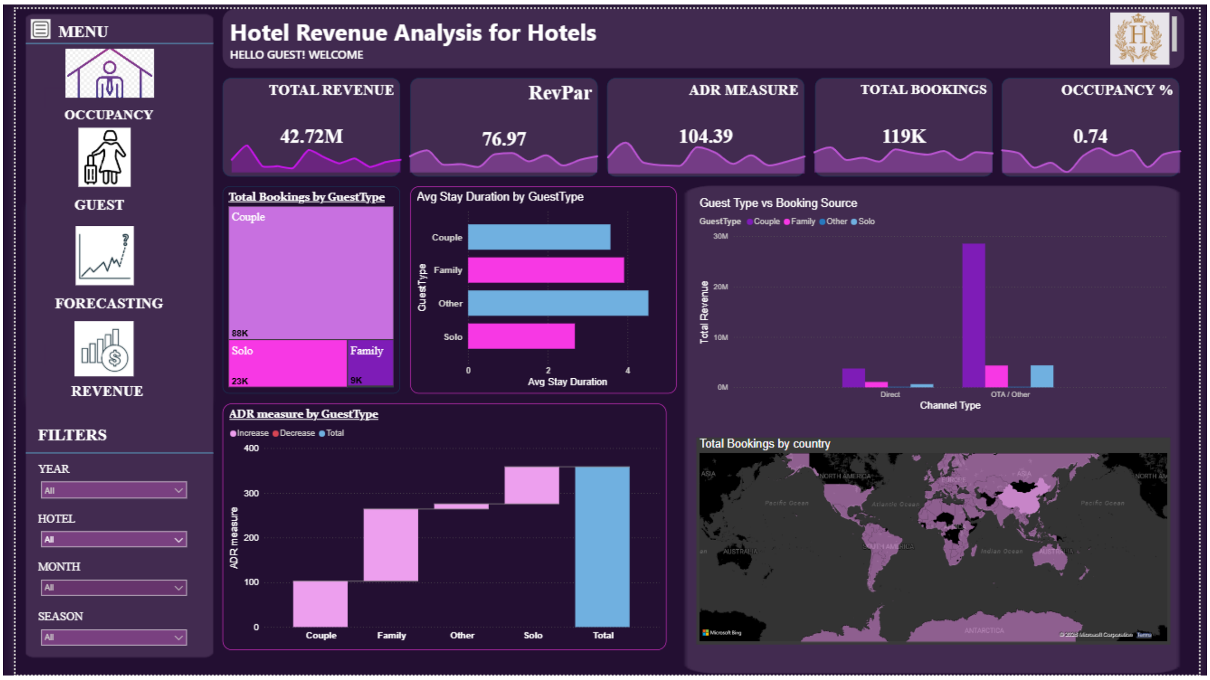


Fig 2 : Guest Analysis Dashboard

• **Forecasting & Cancellation Analysis**

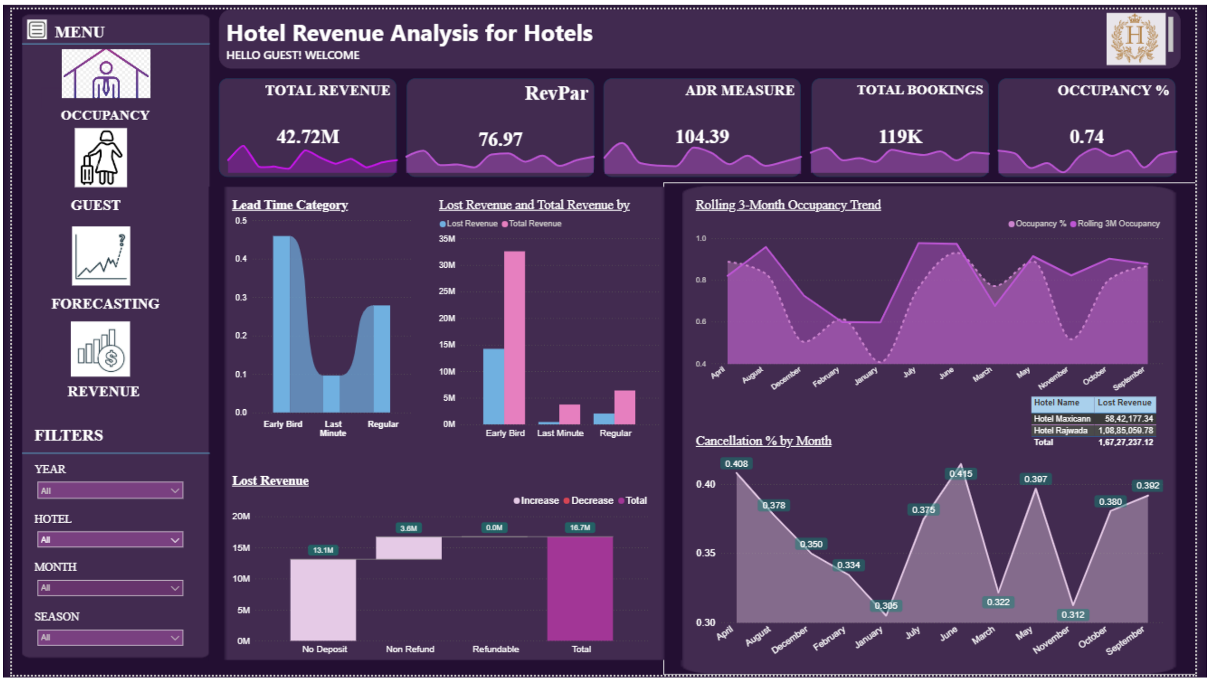


Fig 3: Forecasting and Cancellation Analysis Dashboard

• **Revenue Strategy & Upselling**

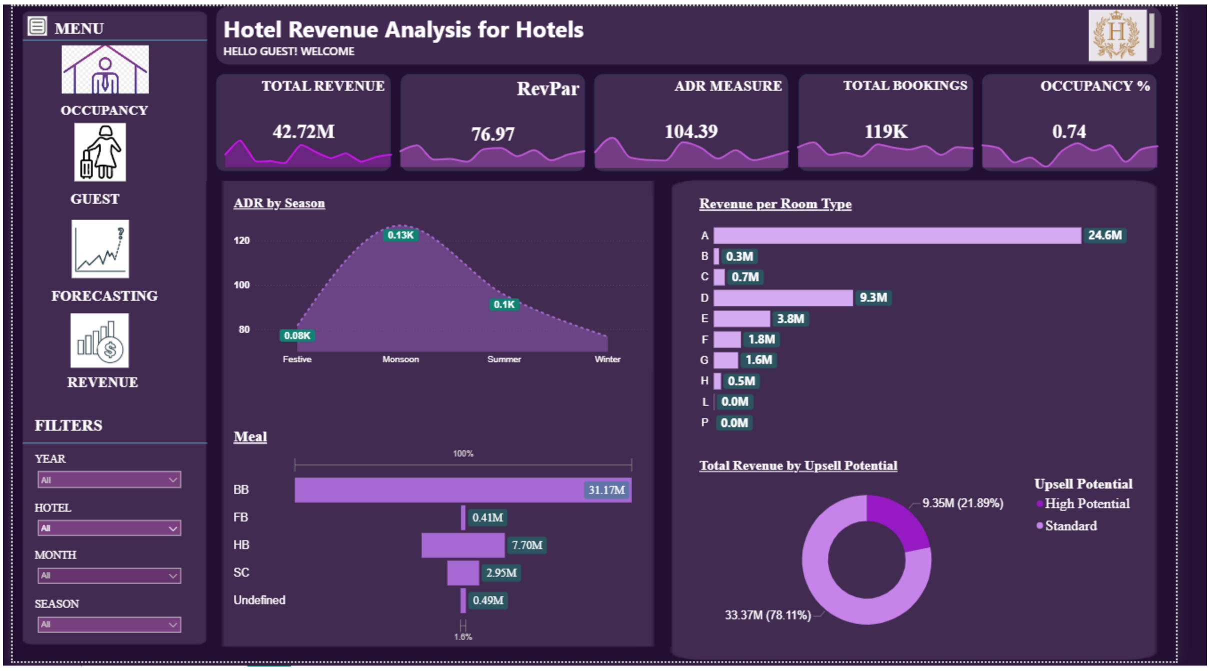


Fig 4: Revenue Strategy Dashboard

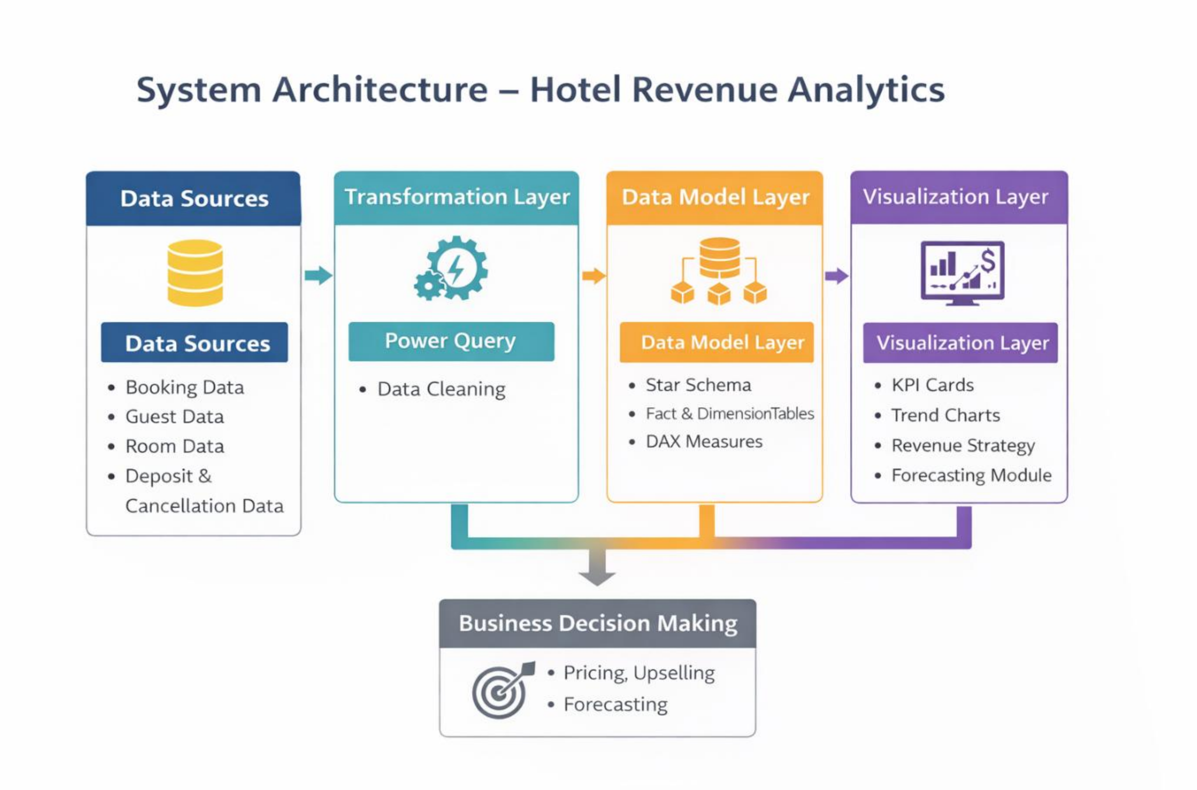
Architecture Diagram

Data Sources: Booking Data, Guest Data, Room Data, Deposit & Cancellation Data

Transformation: Power Query, Data Cleaning

Data Model: Star Schema, Fact & Dimension Tables, DAX Measures

Visualization: KPI Cards, Trend Charts, Revenue Strategy, Forecasting Module.



Conclusion

This project successfully developed a comprehensive Hotel Revenue Analytics Dashboard using Power BI to analyze occupancy patterns, revenue performance, guest behavior, cancellation trends, and pricing effectiveness. By transforming raw booking data into meaningful visual insights, the system enables hotel management to monitor key performance indicators such as Occupancy Percentage, ADR, RevPAR, Total Revenue, Lost Revenue, and Rolling 3-Month Occupancy.

The analysis highlights how revenue performance is influenced not only by the number of bookings but also by seasonal demand, lead time behavior, deposit policies, room categories, meal preferences, and upsell potential. The forecasting and cancellation module helps identify revenue leakage and high-risk booking segments, while the revenue strategy module supports dynamic pricing and upselling decisions.

Overall, the project demonstrates that data-driven decision-making can significantly improve profitability, reduce financial risk, and enhance strategic planning in the hospitality industry. The dashboard serves as a centralized decision-support system that assists management in optimizing revenue without necessarily increasing occupancy.

Future Enhancements

Although the current system provides strong analytical capabilities, it can be further enhanced by integrating advanced predictive and automation features. Future improvements may include incorporating machine learning models to predict cancellation probability and forecast revenue with higher accuracy. Real-time integration with hotel booking systems would allow continuous monitoring of performance metrics and faster managerial response.

The solution can also be extended by developing an automated pricing recommendation engine that dynamically adjusts room rates based on demand patterns, seasonal trends, and occupancy levels. Additionally, incorporating customer lifetime value analysis and repeat guest tracking would help design more personalized marketing and upselling strategies. Expanding the dashboard for mobile access and cross-property benchmarking across hotel chains would further increase its strategic value and scalability.

With these enhancements, the system can evolve from a descriptive analytics platform into a predictive and prescriptive decision-support solution for hotel revenue management.